

Project Proposal: Trajectory-Controlled Robot Arm (Arc-Drawing Plotter)

Abstract

This proposal outlines the design and development of a low-cost, trajectory-controlled robot arm/plotter system capable of executing smooth motion trajectories, such as drawing arcs. Using stepper motors, belts, and an Arduino-based motion controller, the system will demonstrate trajectory planning techniques relevant to robotic manipulators. The project is targeted at completion within eight weeks and aligns with MTech-level standards for robotics and automation.

Background & Objective

Trajectory planning is fundamental to robotics for ensuring smooth and precise motion. Inspired by MathWorks tutorials on quintic polynomial trajectory generation, this project aims to: - Build a low-cost Core-XY based robot arm/plotter. - Implement trajectory generation algorithms (quintic/S-curve). - Evaluate trajectory accuracy using visual error measurement.

System Overview

- **Structure:** Core-XY motion stage with pen holder.
- **Actuation:** 2 × NEMA17 stepper motors with belt drive.
- **Control:** Arduino Uno with CNC/GRBL shield.
- **Sensing/Validation:** Endstops for homing and webcam for path error measurement.
- **Power:** 12V 5A SMPS supply.

Bill of Materials & Cost Analysis

Essential Components: - 2 × NEMA17 stepper motors @ ₹1,054 each = ₹2,108 - 2 × DRV8825 stepper drivers @ ₹94 each = ₹188 - Arduino Uno = ₹1,179 - CNC Shield = ₹163 - GT2 belt & pulley kit = ₹1,775 - Smooth rods & bearings = ₹2,000 - 12V 5A SMPS = ₹1,000 - 3D printed frame parts & acrylic/aluminium support = ₹3,000 - Endstops, wiring, pen mount = ₹800 - Webcam for error measurement = ₹1,500 - Contingency/spares = ₹700

Total Estimated Cost: ₹14,413 (within the ₹15,000–20,000 student-funded range)

Time Schedule (8 Weeks)

- **Week 1–2:** Literature study, CAD design, component procurement.
- **Week 3:** Frame fabrication and assembly.
- **Week 4:** Motor wiring, driver setup, GRBL firmware test.
- **Week 5:** Implement basic motion commands, homing cycles.
- **Week 6:** Integrate trajectory planning (quintic/S-curve).

- **Week 7:** Trial and error testing (arc drawing, path measurement).
- **Week 8:** Accuracy evaluation, documentation, and demo.

Expected Outcomes

- Working Core-XY based plotter capable of drawing arcs and curves.
- Demonstration of trajectory planning algorithms.
- Quantitative error analysis using webcam-based validation.

Risk Factors & Mitigation

- **Stepper skipping steps** → Add current tuning and microstepping.
- **Frame rigidity issues** → Use aluminium supports where possible.
- **Software bugs** → Validate with simulation before hardware runs.

Budget Summary

- Estimated Total: ₹14,413
- Funding: Student-backed
- Scope for $\pm 10\%$ fluctuation due to local procurement

Approval Request

This project represents a feasible and academically significant demonstration of robotic trajectory planning. We request faculty approval to proceed with procurement and development.